

Web Scraping

(Beautiful Soup, Jsoup)

Introduction to CSV Files

A **CSV (Comma-Separated Values)** file is a plain-text file used to store tabular data. Each line represents a record (row), and the values within each record are separated by commas.

CSV files are commonly used to exchange data between databases, spreadsheets, and other applications.

For example, consider a CSV file named students.csv:

```
StudentID,Name,Major,GPA  
1001,Alice Johnson,Computer Science,3.8  
1002,Brian Smith,Biology,3.5  
1003,Carla Davis,Business,3.9  
1004,Daniel Lee,Mathematics,3.2
```

The first row contains the column names, while the remaining rows contain the actual data.

Importing CSV Files into MySQL

Importing is the process of transferring data from an external file into a database table. For example, a university might receive student information in a CSV file and import that information into its MySQL database.

There are two common ways to import CSV files into MySQL:

1. Using the **Table Data Import Wizard** in MySQL Workbench.
2. Using the SQL LOAD DATA statement.

Enabling File Import Into MySQL Workbench

For security reasons, **local file loading may be disabled by default** in MySQL. When attempting to import a CSV file, users may encounter an error such as:
Error Code: 3948.

Loading local data is disabled;

This must be enabled on both the client and server sides.

To resolve this issue, local file loading must be enabled on both:

- 1. MySQL Server** – The database server must permit local file loading.
- 2. MySQL Workbench** – The client connection must permit local file loading.

-- In MySQL Script,

```
-- SHOW GLOBAL VARIABLES LIKE 'local_infile';  
-- SET GLOBAL local_infile = 1;
```

-- In MySQL Workbench,

```
-- Go back to the MySQL Workbench home screen.  
-- Find your connection.  
-- Right-click it and choose Edit Connection.  
-- Click Advanced.  
-- In the Others: box, add:  
-- OPT_LOCAL_INFILE=1
```

Importing the CSV File Using SQL LOAD DATA

Import the CSV File

Suppose the file is located at:

`C:/Users/joann/Downloads/Student.csv`

Execute:

```
LOAD DATA LOCAL INFILE 'C:/Users/joann/Downloads/Student.csv'  
INTO TABLE Student  
FIELDS TERMINATED BY ','  
OPTIONALLY ENCLOSED BY '"'  
LINES TERMINATED BY '\r\n'  
IGNORE 1 ROWS;
```

How to Handle NULL Values

One important consideration when importing CSV files is how missing values are handled.

Suppose a CSV file contains:

StudentID,Name,Major,GPA

1001,Alice Johnson,Computer Science,3.8

1002,Brian Smith,,3.5

1003,Carla Davis,Business,3.9

Notice that Brian Smith has no value for Major.

An empty field in a CSV file does not always automatically become an SQL NULL.

Depending on the import method, it might be imported as an empty string (") instead.

NULL vs. Empty String

These values have different meanings in SQL.

Value	Meaning
NULL	Missing or unknown value
' '	An empty string
'NULL'	A string containing the word NULL

CSV Files and Double Quotation Marks

CSV files sometimes contain values enclosed in double quotation marks.

For example:

StudentID,Name,Major,GPA

1001,"Alice Johnson","Computer Science",3.8

1002,"Brian Smith","Biology",3.5

Double quotation marks are commonly used as **text qualifiers** in CSV files. They are particularly useful when a value contains a comma.

Exporting Data from MySQL to CSV

Exporting is the process of transferring data from a database table into an external file. For example, a university might export student information into a CSV file to share with another department or analyze using Microsoft Excel.

There are two common ways to export MySQL data:

1. Using MySQL Workbench's graphical interface.
2. Using the SQL `SELECT ... INTO OUTFILE` statement.

What Is Beautiful Soup?

Beautiful Soup is a Python library used to extract information from HTML and XML documents. It is commonly used in **web scraping**, the process of automatically collecting information from websites.

Websites typically display information using HTML (HyperText Markup Language). Beautiful Soup allows Python programs to examine the HTML structure, locate specific elements, and extract information such as text, links, product names, prices, and other data.

For example, suppose an online bookstore displays the following information:

Book Title	Price	Availability
Python Programming	\$39.99	In Stock
Database Systems	\$59.99	In Stock
Introduction to Java	\$44.99	Out of Stock

A Python program using Beautiful Soup can automatically extract this information instead of requiring someone to manually copy it from the website.

The collected information can then be stored in a database such as MySQL.